

This Java 8 challenge tests your knowledge of [Lambda expressions](#)!

Write the following methods that return a lambda expression performing a specified action:

1. PerformOperation isOdd(): The lambda expression must return *true* if a number is odd or *false* if it is even.
2. PerformOperation isPrime(): The lambda expression must return *true* if a number is prime or *false* if it is composite.
3. PerformOperation isPalindrome(): The lambda expression must return *true* if a number is a palindrome or *false* if it is not.

Input Format

Input is handled for you by the locked stub code in your editor.

Output Format

The locked stub code in your editor will print *T* lines of output.

Sample Input

The first line contains an integer, *T* (the number of test cases).

The *T* subsequent lines each describe a test case in the form of 2 space-separated integers:

The first integer specifies the condition to check for (1 for Odd/Even, 2 for Prime, or 3 for Palindrome). The second integer denotes the number to be checked.

```
5
1 4
2 5
3 898
1 3
2 12
```

Sample Output

```
EVEN
PRIME
PALINDROME
ODD
COMPOSITE
```

```

1  import java.util.*;
2
3  interface PerformOperation {
4      boolean check(int a);
5  }
6
7  public class Solution {
8
9      public static PerformOperation isOdd() {
10         return new PerformOperation() {
11             public boolean check(int n) {
12                 return n % 2 != 0;
13             }
14         };
15     }
16
17     public static PerformOperation isPrime() {
18         return new PerformOperation() {
19             public boolean check(int n) {
20                 if (n < 2) {
21                     return false;
22                 }
23
24                 for (int i = 2; i * i <= n; i++) {
25                     if (n % i == 0) {
26                         return false;
27                     }
28                 }
29
30                 return true;
31             }
32         };
33     }
34
35     public static PerformOperation isPalindrome() {
36         return new PerformOperation() {
37             public boolean check(int n) {
38                 int original = n;
39                 int reversed = 0;
40
41                 while (n > 0) {
42                     reversed = reversed * 10 + n % 10;
43                     n /= 10;
44                 }
45
46                 return original == reversed;
47             }
48         };
49     }
50
51     public static void main(String[] args) {
52         Scanner sc = new Scanner(System.in);
53
54         int t = sc.nextInt();
55
56         while (t-- > 0) {
57             int condition = sc.nextInt();
58             int number = sc.nextInt();
59
60             if (condition == 1) {
61                 System.out.println(
62                     isOdd().check(number) ? "ODD" : "EVEN"
63                 );
64             }
65             else if (condition == 2) {
66                 System.out.println(
67                     isPrime().check(number) ? "PRIME" : "COMPOSITE"
68                 );
69             }
70             else if (condition == 3) {
71                 System.out.println(
72                     isPalindrome().check(number)
73                     ? "PALINDROME"
74                     : "NOT PALINDROME"
75                 );
76             }
77         }
78
79         sc.close();
80     }
81 }
82

```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

[Download](#)

```
1 5
2 1 4
3 2 5
4 3 898
5 1 3
6 2 12
```

Your Output (stdout)

```
1 EVEN
2 PRIME
3 PALINDROME
```